

Assignment and Paper-Discussion



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Discussion of home assignment



Open Questions and Problems?

Example: Monopoly

- The demand function in the market for a good is **$p = 500 - 3x$** (p = price, x = quantity).
- This is offered by only one firm (**supply monopoly**). The marginal costs of production are **$MC = 20 + 4x$** .
- The state imposes a **quantity tax** with a tax amount of **$t = 20$** on the good, which must be paid by the producers.
- **Notation:**
- **p^*** -> Equilibrium price without tax
- **p'** -> new market price
- **p_n** -> Price after tax

a) Calculate the quantity sold and the price at equilibrium before and after the introduction of the tax. State the equilibrium price without tax, the new market price, and the price after tax.

■ **Before Tax**

■ $E = p \cdot x = (500 - 3x) \cdot x = 500x - 3x^2$

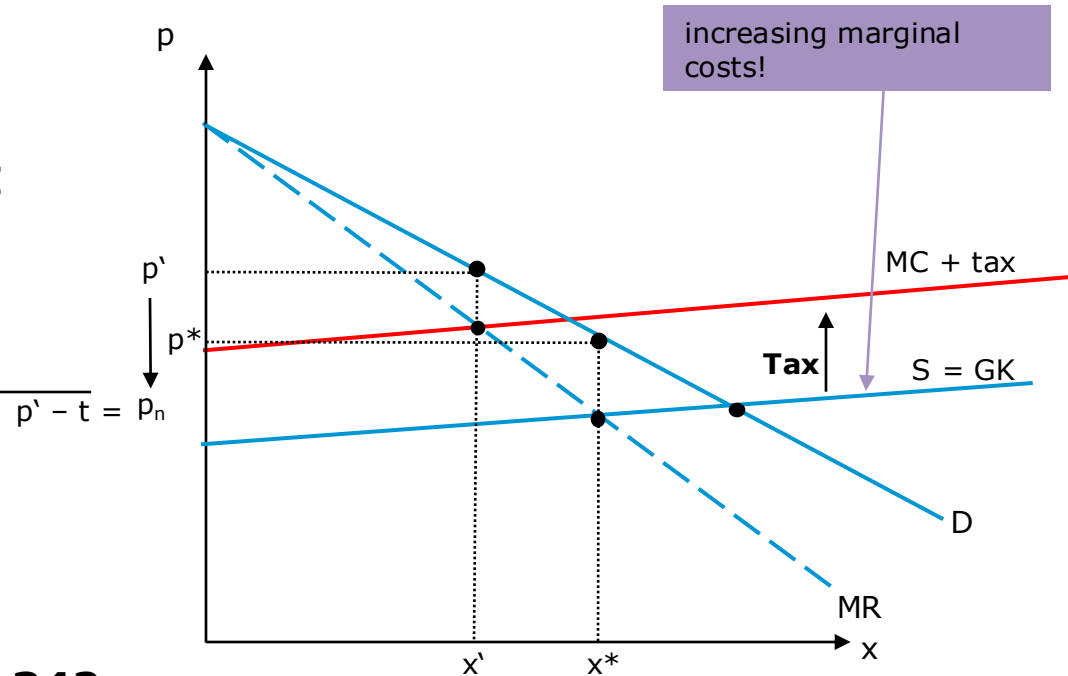
■ $MR = 500 - 6x$

■ Equilibrium Condition: $MR = MC$
 $500 - 6x = 20 + 4x$

■ $x^* = 48 \rightarrow p^* = 356$

■ After Tax: $MR = MC + t$
 $500 - 6x = 20 + 4x + 20$

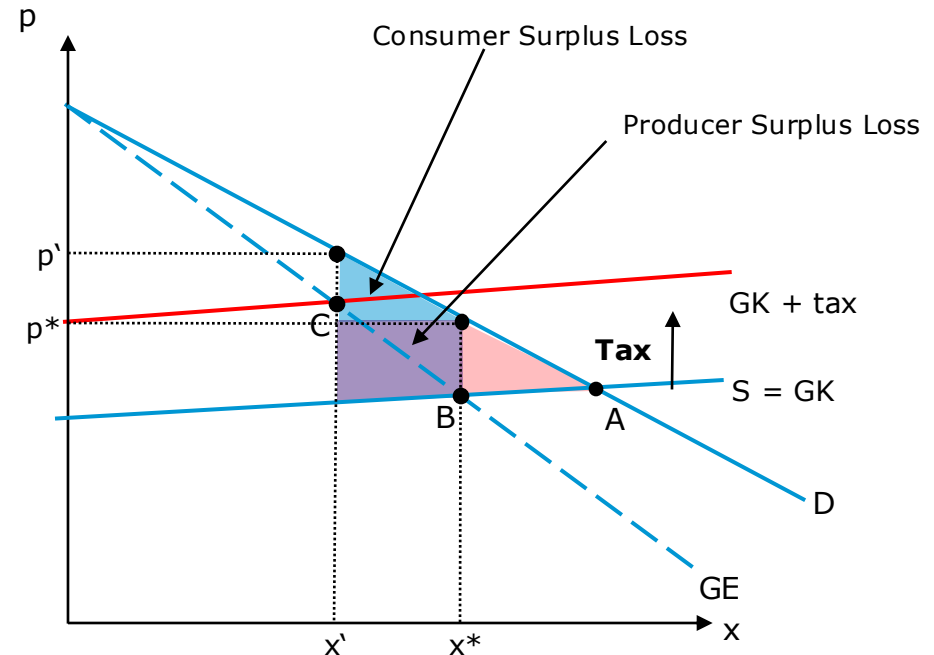
■ $x' = 46 \rightarrow p' = 362$



Price after Tax: $p_n = 362 - 20 = 342$

b) Calculate the total welfare loss caused by the tax and what proportion of the welfare loss falls on the consumers and the producers.

- Change in the quantity traded → lost surplus
- **Consumers**
- $(p' - p^*) \cdot (x^* - x') / 2 = 6$
- → **2%**
- **Producers**
- $(x^* - x') \cdot (p^* - MC(x^*)) + (x^* - x') \cdot (MC(x^*) - MC(x')) / 2 = 296$
- **98%**



Not the same as tax burden distribution

Paper-Discussion



Benzarti et al (2024): *"Can VAT Cuts and Anti-Profiteering Measures Dampen the Effects of Food Price Inflation?"* NBER Working Paper No. 32241. R&R American Economic Review.

Overview:

- **Overview:**
- What is the research question to be answered in the paper?
- Which policies are being analyzed?
- What goals were pursued with the policies?

Overview:

- *Benzarti et al (2024): "Can VAT Cuts and Anti-Profiteering Measures Dampen the Effects of Food Price Inflation?" NBER Working Paper No. 32241. R&R American Economic Review.*
- **Research Question:** Is a VAT cut a helpful measure against inflation?
- Argentina: High inflation, strong fall of the Peso, elections
- 21% VAT cut from August to December 2019 on selected basic food products
- Anti-Profiteering Measures: after the VAT cut is lifted, there are price caps for certain products
- Goal: Curb inflation

Primary
Elections
August 11

August 12
Argentine peso
drops 30%

0% VAT on 13 categories of basic food basket
21% VAT on remaining categories

August 15
0% VAT
announced

December 31
End of VAT cut
+ caps announced

Timeline
2019

Questions in small groups: Discuss the questions and write down the most important points on the board.

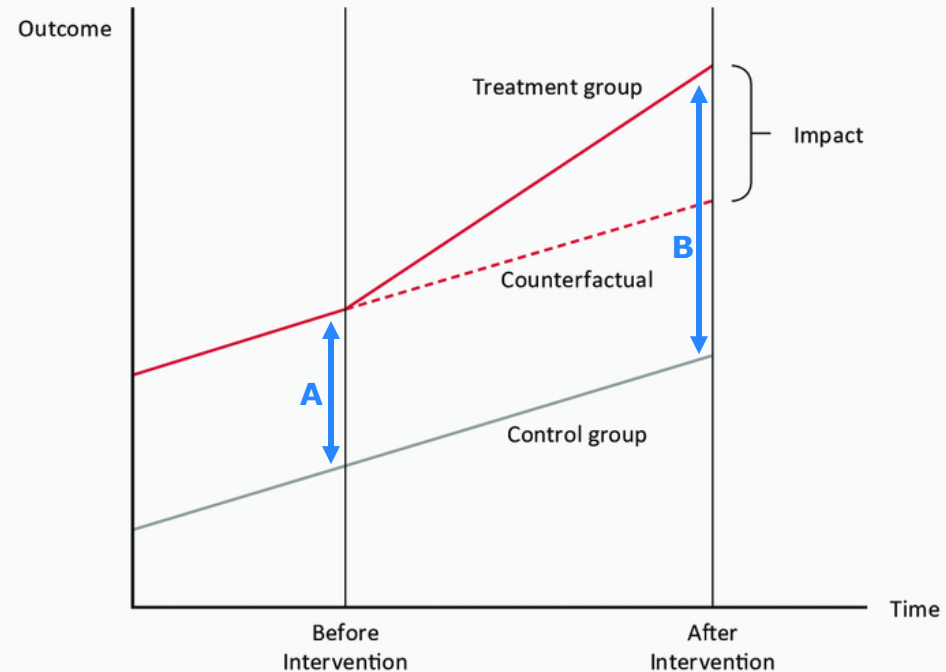
- 1) What is the Difference-in-Difference method and what does it assume? How are the Treatment and Control groups formed? Describe Figure 1a and Table 1.
- 2) What are the main results of the paper? Discuss Figure 1b and 2b. What differences do we see between supermarket chains and independent stores?
- 3) What effect does the price cap have? Figure 4b and 5a. →

Presentation of your discussion results in the large group & joint discussion

- 1) What is the Difference-in-Difference method and what does it assume?
How are the Treatment and Control groups formed? Describe Figure 1a and Table 1.

Difference in Differences Method

- **Intuition**
- We observe 2 groups that behaved similarly before a change → "parallel trends"
- Then there is a change for one group → "event/intervention/policy change"
- We compare both groups before and after the change → $B - A$ gives us the effect
- This is how we model a counterfactual situation
 - We assume that both groups would have followed the same trend without the change (not level)



- Treatment Group 1: those 13 basic food products on which taxes were lowered
- Treatment Group 2: those products on which price caps were imposed after the re-introduction of the tax
- Treatment follows consumption habits in Argentina
- Control group composed of products on which the tax was not lowered: coffee prices as a control group for tea, olive oil for sunflower oil,...
- Treatment & Control products are part of the basic food basket
- Prices in supermarket chains were controlled by the state, but not in independent stores

Table 1: Classification of data into treatment and control

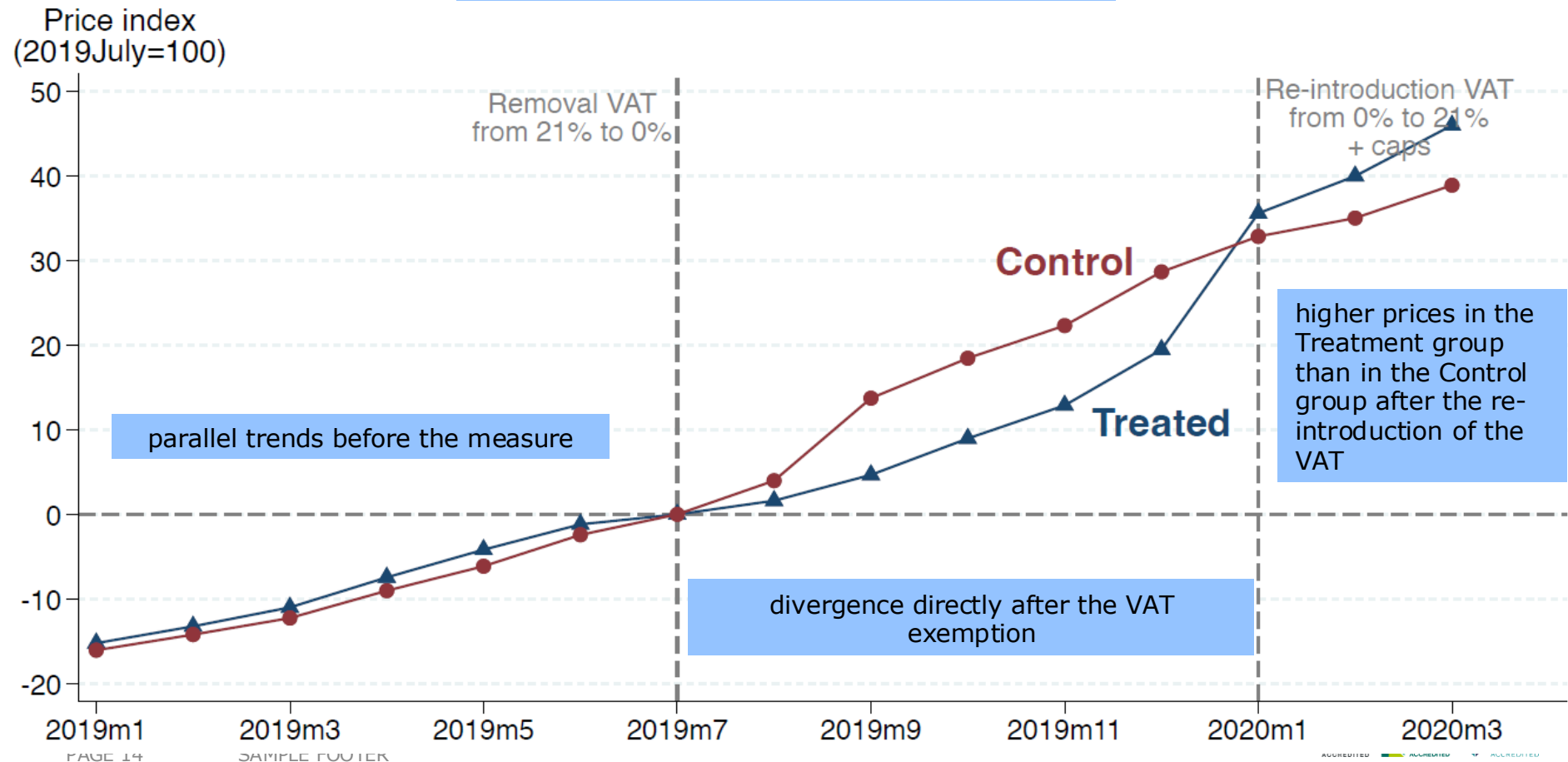
Treatment Temporary 0% VAT	Control Standard 21% VAT
Categories	Categories
Cooking oils (sunflower, corn, mix) Rice Dried pasta Tea, Yerba Mate, and Mate Cocido Sugar Canned vegetables and beans Canned fruits Corn flour (<i>polenta</i>) Wheat flour Fluid milk (whole/skim) Yogurt (whole or skim) Eggs Bread Breadcrumbs and/or batter	Other cooking oils (olive, soy, canola) Rice-based meals Breakfast cereal Coffee Salt Herbs, Spices, & Seasonings Dulce de leche (caramel) Jam and Jelly Other flours Crackers, Biscuits, Toasts, Puddings Chocolate Mayonnaise Vinegar Dried legumes and beans

Notes: This table shows the split of our data into treatment and control categories. Wheat flour and Bread are taxed at the reduced rate of 10.5%. Source: Treatment categories are determined based on Decree 567/2019–Annex. Control products include the remaining categories in our data.

Figure 1: Price Effect of the Temporary VAT cut *without* Anti-Profitteering Measures

(a) Unconditional Means (no caps)

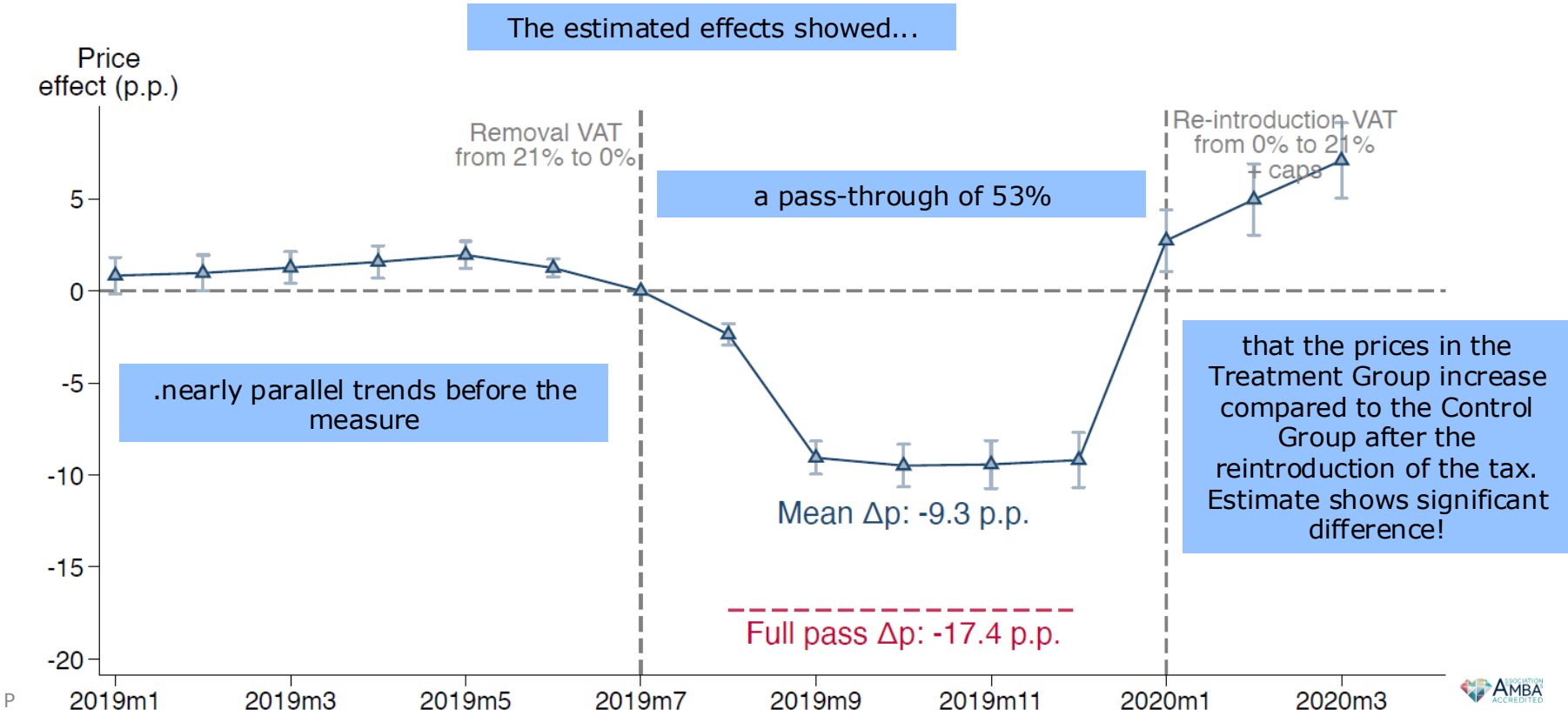
The average price index shows...



- What are the main results of the paper? Discuss Figure 1b and 2b.

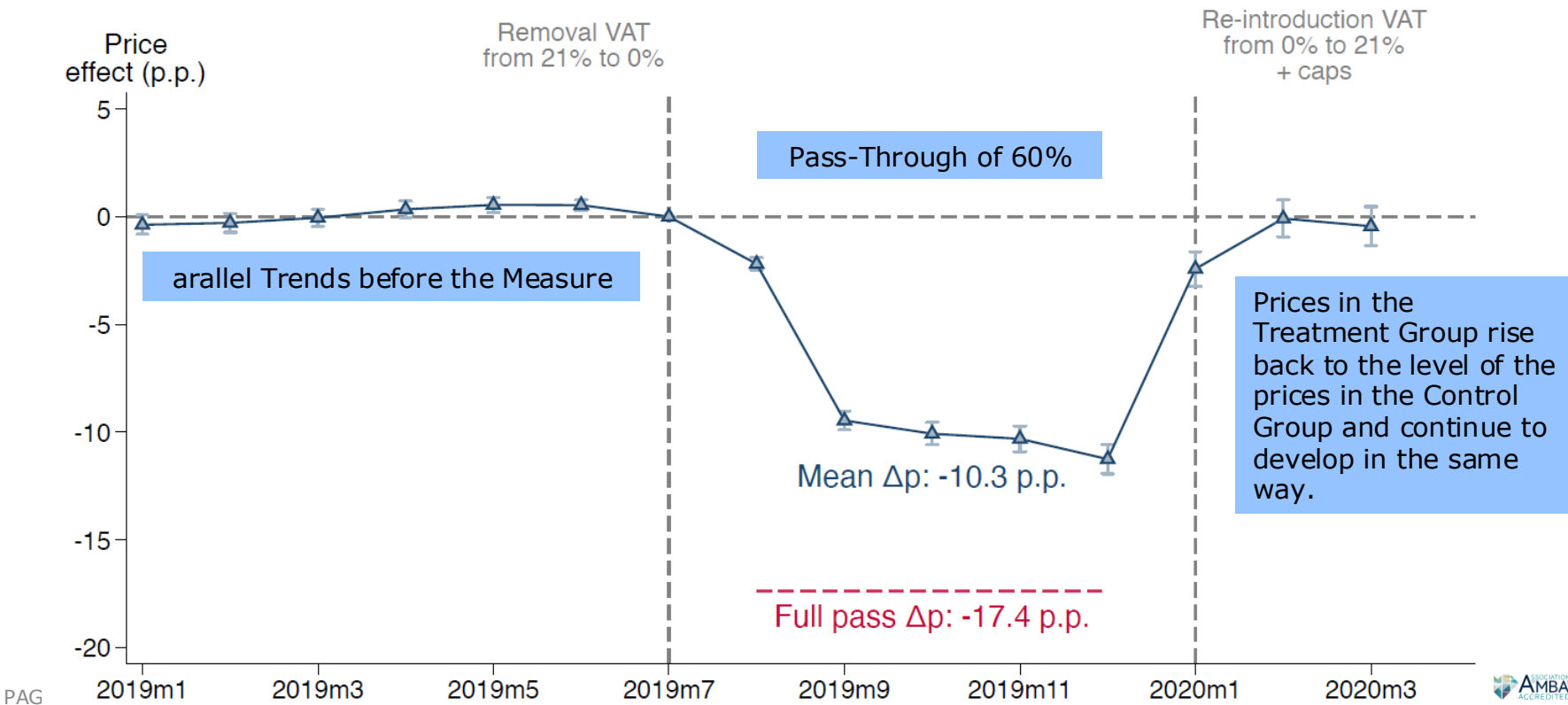
Tax Cut & Increase without Anti-Profiteering Measures

Figure 1: Price Effect of the Temporary VAT cut *without* Anti-Profiteering Measures
(b) Difference-in-Differences Estimates (no caps)



Tax Cut & Increase with Price Cap

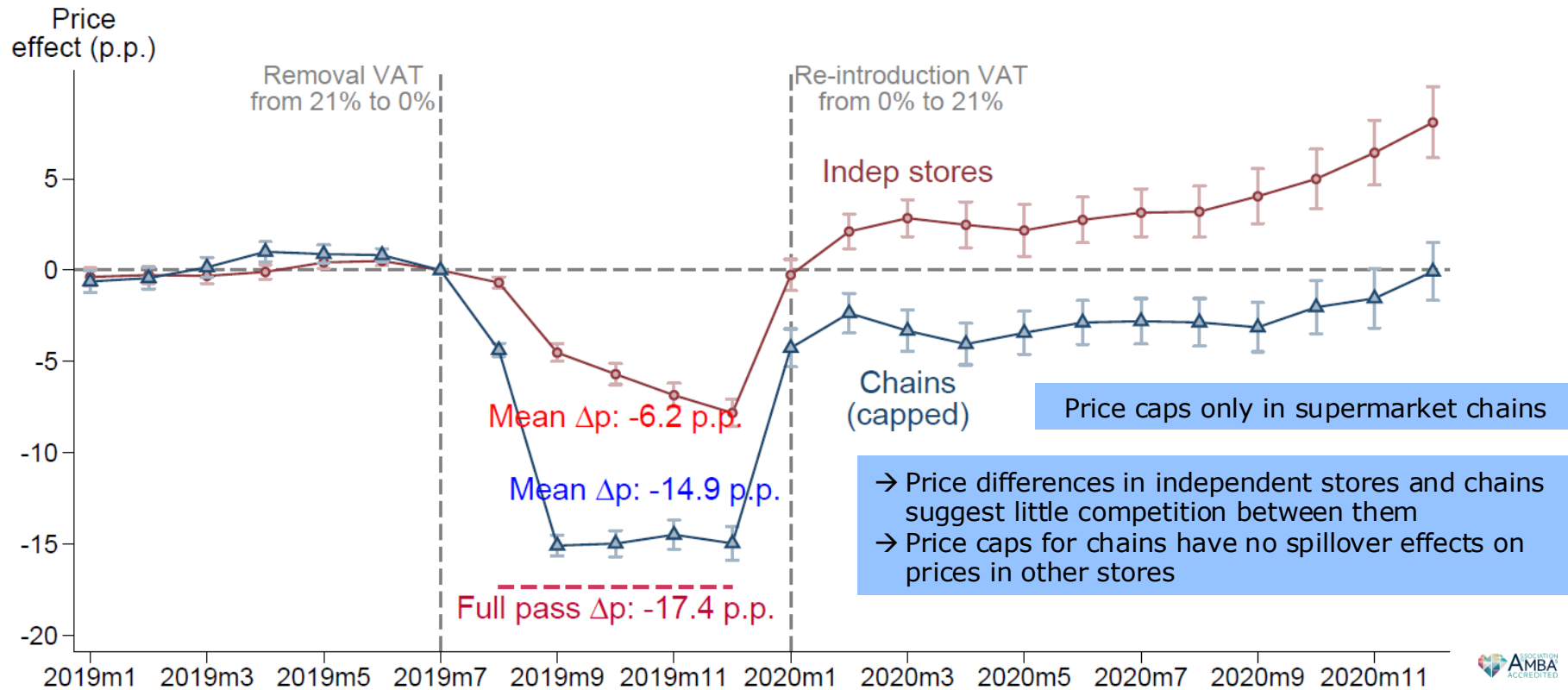
Figure 2: Price Effects of the Temporary VAT cut *With* Anti-Profiteering Measures
(b) Difference-in-Differences Estimates (caps)



- What differences do we see between supermarket chains and independent stores? What effect does the price cap have? Figure 4b and 5a.

Supermarket Chains & Independent Stores

Figure 4: Price Effect of the Temporary VAT Cut in Chain and Independent Supermarkets
(b) Chains and Independent stores (monthly data)



Effect of the price caps in chains

Figure 5: Price Effect of the Anti-Profiteering Measures in Chain Supermarkets

(a) 7% cap versus no cap

